

12. Functions & Methods



-  **Introduction to Functions**
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-  **Syntax - Function Code Demo**
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-  **Syntax with Parameters**
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-  **Parameters v/s Arguments**
✓ video 01:54
-  **What happens in memory? (Call Stack)**
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-  **Call by Value (in Java)**
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-  **Find Binomial Coefficient**
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-  **Inbuilt v/s User defined Methods**
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-  **Function Overloading**
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Functions/Methods



Functions/Methods

↓
block of code | reusable

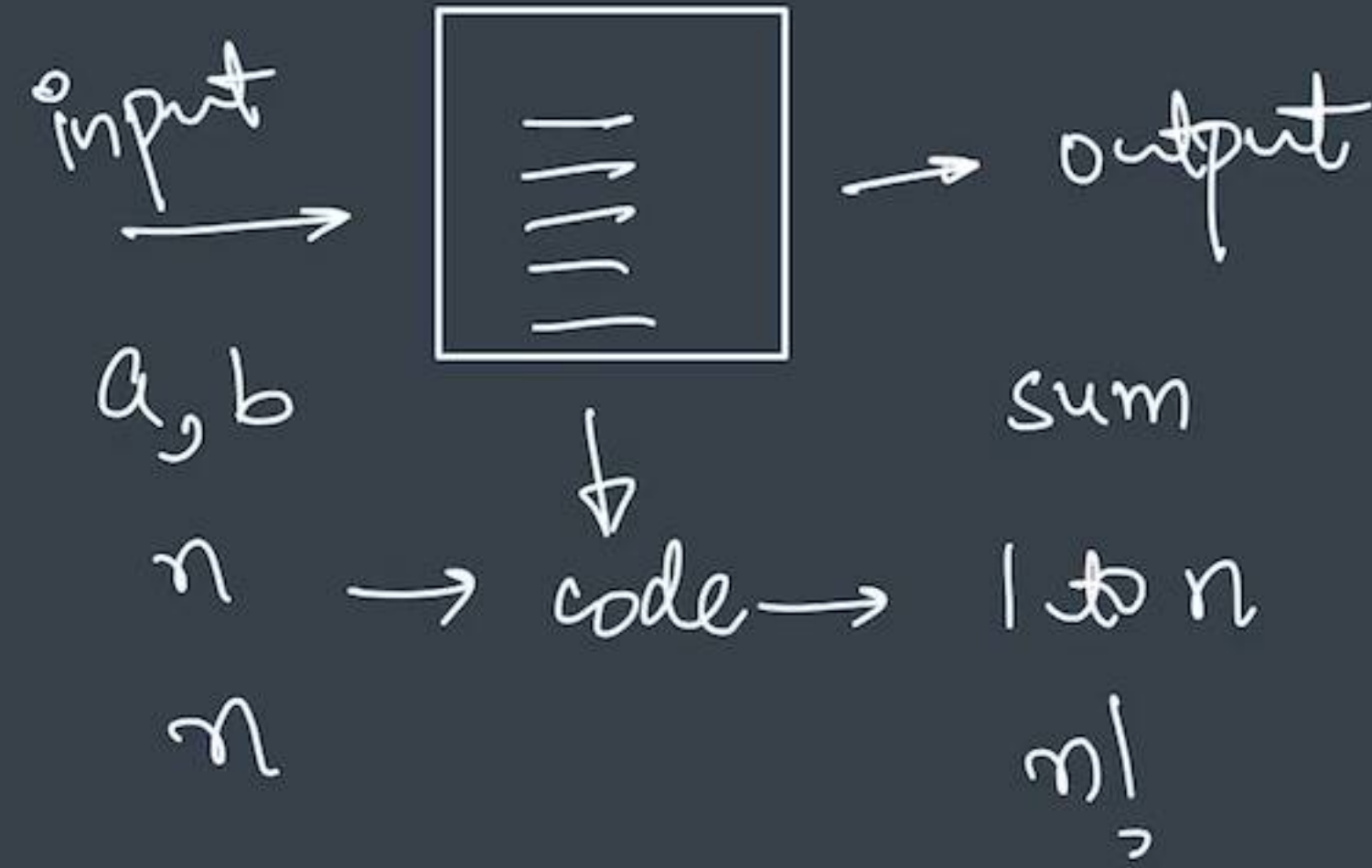
{
}

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Syntax

```
returnType name () {  
    //body  
    return statement;  
}
```



JavaBasics.java > JavaBasics > printHelloWorld()

```
1  public class JavaBasics {
2      public static void printHelloWorld() {
3          System.out.println("Hello World");
4          System.out.println("Hello World");
5          System.out.println("Hello World");
6      }
7
8      public static void main(String args[]) {
9          printHelloWorld(); //function call
10     }
11 }
```

Run | Debug

DEBUG CONSOLE PROBLEMS OUTPUT TERMINAL

```
shradhakhapra@Shradhas-Air Java Programming Language % java JavaBasics.java
shradhakhapra@Shradhas-Air Java Programming Language % java JavaBasics.java
Hello World
shradhakhapra@Shradhas-Air Java Programming Language % java JavaBasics.java
Hello World
Hello World
Hello World
```



JavaBasics.java > JavaBasics > printHelloWorld()

```
1 public class JavaBasics {
2     public static int printHelloWorld() {
3         System.out.println("Hello World");
4         System.out.println("Hello World");
5         System.out.println("Hello World");
6         return 3;
7     }
8
9     public static void main(String args[]) {
10         printHelloWorld(); //function call
11     }
12 }
```

Run | Debug

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DEBUG CONSOLE PROBLEMS OUTPUT TERMINAL

```
shradhakhapra@Shradhas-Air Java Programming Language % java JavaBasics.java
shradhakhapra@Shradhas-Air Java Programming Language % java JavaBasics.java
Hello World
shradhakhapra@Shradhas-Air Java Programming Language % java JavaBasics.java
Hello World
Hello World
Hello World
```



JavaBasics.java > JavaBasics > printHelloWorld()

```
1 public class JavaBasics {  
2     public static int printHelloWorld() {  
3         System.out.pri  
4         System.out.pri  
5         System.out.pri  
6         //return 3;  
7     }
```

int JavaBasics.printHelloWorld()

This method must return a result of type int Java(603979884)

[View Problem](#) [Quick Fix... \(⌘.\)](#)

Run | Debug

```
8 public static void main(String args[]) {  
9     printHelloWorld(); //function call  
10 }  
11 }  
12
```

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DEBUG CONSOLE

PROBLEMS

1

OUTPUT

TERMINAL

```
shradhakhapra@Shradhas-Air Java Programming Language % java JavaBasics.java  
shradhakhapra@Shradhas-Air Java Programming Language % java JavaBasics.java  
Hello World  
shradhakhapra@Shradhas-Air Java Programming Language % java JavaBasics.java  
Hello World  
Hello World  
Hello World
```



Syntax with Parameters

```
returnType name ( type param1, type param2) {  
    //body  
    return statement;  
}
```



JavaBasics.java > JavaBasics > calculateSum(int, int)

```
6      System.out.println("Hello World");
7  }
8
9  public static int calculateSum(int num1, int num2) {
10      int sum = num1 + num2;
11      return sum;
12  }
13
```

Run | Debug

```
14 public static void main(String args[]) {
15     Scanner sc = new Scanner(System.in);
16     int a = sc.nextInt();
17     int b = sc.nextInt();
18     int sum = calculateSum(a, b);
19     System.out.println("sum is : " + sum);
20 }
```

DEBUG CONSOLE PROBLEMS 1 OUTPUT TERMINAL

```
shradhakhapra@Shradhas-Air Java Programming Language % java JavaBasics.java
5
10
sum is : 15
shradhakhapra@Shradhas-Air Java Programming Language % java JavaBasics.java
5
10
```



JavaBasics.java > JavaBasics > main(String[])

```
5      System.out.println("Hello World");  
6      System.out.println("Hello World");  
7  }
```

```
9  public static int calculateSum(int num1, int num2) { //parameters or formal parameters  
10     int sum = num1 + num2;  
11     return sum;  
12 }
```

Run | Debug

```
14 public static void main(String args[]) {  
15     Scanner sc = new Scanner(System.in);  
16     int a = sc.nextInt();  
17     int b = sc.nextInt();  
18     int sum = calculateSum(a, b); // arguments or actual parameters  
19     System.out.println("sum is : " + sum);  
20 }  
21 }  
22
```



Syntax with Parameters

```
returnType name ( type param1, type param2) {  
    //body  
    return statement;  
}
```

✓ formal params
params
↓
definition

actual params ✓
arguments
↓
call



What happens in Memory?

Call stack

Recursion ✓

```
ps void sum(int a, int b)
{
    int sum = a + b;
}
```

```
ps void main(String arg[])
{
    sum()
}
```



call stack



What happens in Memory?

Call stack

Recursion ✓

```
ps void sum(int a, int b)
{
    int sum = a + b;
    return;
}

ps void main(String arg[])
{
    sum();
}
```



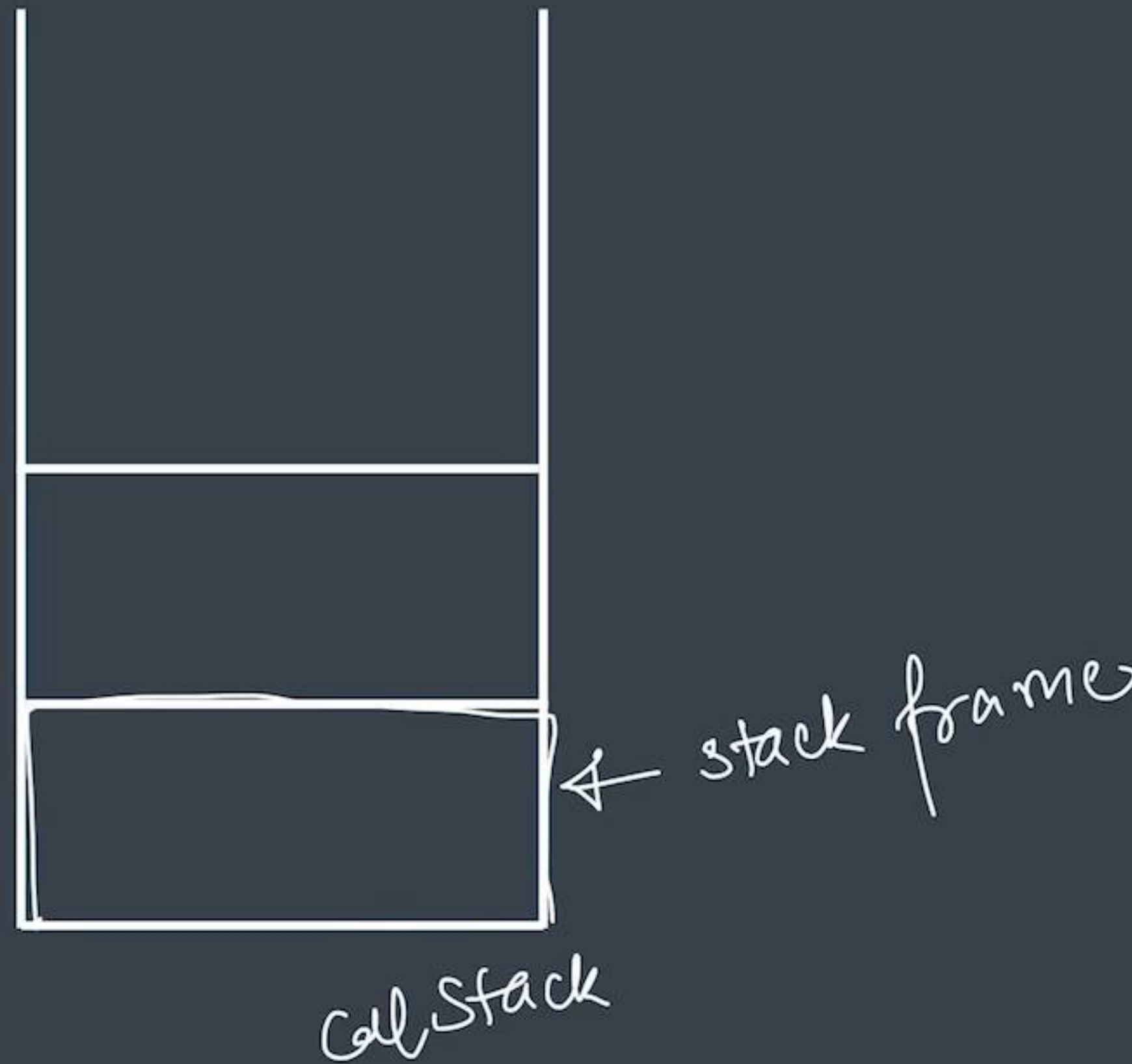
What happens in Memory?

Call stack

Recursion ✓

```
ps void sum(int a, int b)
{
    int sum = a + b;
    return;
}

ps void main(String arg[])
{
    sum();
}
```



JavaBasics.java > JavaBasics > swap(int, int)

```
14 public static void swap(int a, int b) {  
15     //swap  
16     int temp = a;  
17     a = b;  
18     b = temp;  
19 }  
20  
21
```

Run | Debug

```
22 public static void main(String args[]) {  
23     //swap - values exchange  
24     int a = 5;  
25     int b = 10;  
26     swap(a, b);  
27     System.out.println("a = " + a);  
28     System.out.println("b = " + b);  
29 }
```

DEBUG CONSOLE

PROBLEMS

1

OUTPUT

TERMINAL

a = 10

b = 5

shradhakhapra@Shradhas-Air Java Programming Language % java JavaBasics.java

a = 5



Call by Value

Java always calls by value

```
changeA(int a) {  
    a = 10;  
}  
  
psvm() {  
    int a = 5;  
    changeA(a);  
    sysout(a);  
}
```

changeA	a = 5 10
main	a = 5

Call by Reference



Call by Value

Java always calls by value

```
changeA(int a) {  
    a = 10;  
}  
  
public {  
    int a = 5;  
    changeA(a);  
    sysout(a);  
}
```

copy

Call by Reference



Call by Value

Java always calls by value

```
changeA(int a) {
```

```
    a = 10;
```

```
}
```

```
public {
```

```
    int a = 5;
```

```
    changeA(a);
```

```
    sysout(a);
```

```
}
```

JAVA Programming Language

Call by Reference

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Find Product of a & b $\underline{a = 3}, b = 5$

PS `int multiply(int a, int b) {`
 `int product = a * b;`
 `return product;`
 `}`

`int prod = multiply(3, 5);`
`print(prod)`



```
22 public static int multiply(int a, int b) {  
23     int product = a * b;  
24     return product;  
25 }  
26
```

Run | Debug

```
27 public static void main(String args[]) {  
28     int a = 3;  
29     int b = 5;  
30     int prod = multiply(a, b);  
31     System.out.println("a * b = " + prod);  
32 }  
33 }  
34
```

DEBUG CONSOLE

PROBLEMS

1

OUTPUT

TERMINAL

```
shradhakhapra@Shradhas-Air Java Programming Language % java JavaBasics.java
```

```
a * b = 15
```

```
shradhakhapra@Shradhas-Air Java Programming Language %
```



```
24     return product;
25 }
26
Run | Debug
27 public static void main(String args[]) {
28     int a = 3;
29     int b = 5;
30     int prod = multiply(a, b);
31     System.out.println("a * b = " + prod);
32     prod = multiply(10, 20);
33     System.out.println("a * b = " + prod);
34 }
35 }
36
```

DEBUG CONSOLE

PROBLEMS

1

OUTPUT

TERMINAL

```
shradhakhapra@Shradhas-Air Java Programming Language % java JavaBasics.java
a * b = 15
shradhakhapra@Shradhas-Air Java Programming Language % java JavaBasics.java
a * b = 15
a * b = 200
shradhakhapra@Shradhas-Air Java Programming Language %
```



Factorial of a number, n $n = 4$

$$n! = n \times (n-1) \times (n-2) \times (n-3) \dots \times 1 = n!$$

$$4! = 4 \times 3 \times 2 \times 1$$

$$3! = 3 \times 2 \times 1$$

$$2! = 2 \times 1$$

$$1! = 1$$

$$0! = 1$$

True



Factorial of a number, n $n = 4$

$n!$

```
int f = 1;  
for (i = 1 to n) {  
    f = f * i;  
}  
return f; ← factorial
```



Factorial of a number, n

n!

n = 4

sum =

```
int f = 1;
```

```
for (i = 1 to n) {
```

```
    f = f * i;
```

```
}
```

```
return f; ← factorial
```

$24 = 4! = 4 \times 3 \times 2 \times 1$

$f = \underline{1} \times \cancel{2} \times \cancel{3} \times \underline{24}$

i = 1

$f = 1 \times 1 = 1$

i = 2

$f = 1 \times 2 = \underline{2}$

i = 3

$f = 2 \times 3 = \underline{6}$

i = 4

$f = 6 \times 4 = \underline{24}$

```
20 }  
21  
22 public static int multiply(int a, int b) {  
23     int product = a * b;  
24     return product;  
25 }
```

```
26  
27 public static int factorial(int n) {  
28     int f = 1;  
29  
30     for(int i=1; i<=n; i++) {  
31         f = f * i;  
32     }  
33  
34     return f; //factorial of n  
35 }  
36
```

Run | Debug

```
37 public static void main(String args[]) {  
38     System.out.println(factorial(4));  
39 }  
40 }  
41
```



```
28     int f = 1;
29
30     for(int i=1; i<=n; i++) {
31         f = f * i;
32     }
33
34     return f; //factorial of n
35 }
36
Run | Debug
37 public static void main(String args[]) {
38     System.out.println(factorial(7));
39 }
40 }
```

DEBUG CONSOLE

PROBLEMS

1

OUTPUT

TERMINAL

```
shradhakhapra@Shradhas-Air Java Programming Language % java JavaBasics.java
24
shradhakhapra@Shradhas-Air Java Programming Language % java JavaBasics.java
5040
shradhakhapra@Shradhas-Air Java Programming Language %
```



Binomial Coefficient

$$n = \overset{5}{\cancel{10}}, r = 2$$

factorial(n)

Reusable
= APNA
COLLEGE

$${}^n C_r = \frac{n!}{r! (n-r)!}$$



Binomial Coefficient

$$n = \overset{5}{\cancel{10}}, r = 2$$

factorial(n)

Renegade
= APNA
COLLEGE

$${}^n C_r = \frac{n!}{r! (n-r)!}$$

```
int binCoeff (int n, int r) {  
    ✓ n_fact = factorial(n); (a)  
    ✓ r_fact = factorial(r); (b)  
    ✓ nmr_fact = factorial(n-r); (c)  
}
```

$$\bigcirc = a / (b * c);$$



```
36  
37 public static int binCoeff(int n, int r) {  
38     int fact_n = factorial(n);  
39     int fact_r = factorial(r);  
40     int fact_nmr = factorial(n-r);  
41  
42     int binCoeff = fact_n / (fact_r * fact_nmr);  
43     return binCoeff;  
44 }
```

Run | Debug

```
45 public static void main(String args[]) {  
46     System.out.println(binCoeff(5, 2));  
47 }
```

DEBUG CONSOLE

PROBLEMS

1

OUTPUT

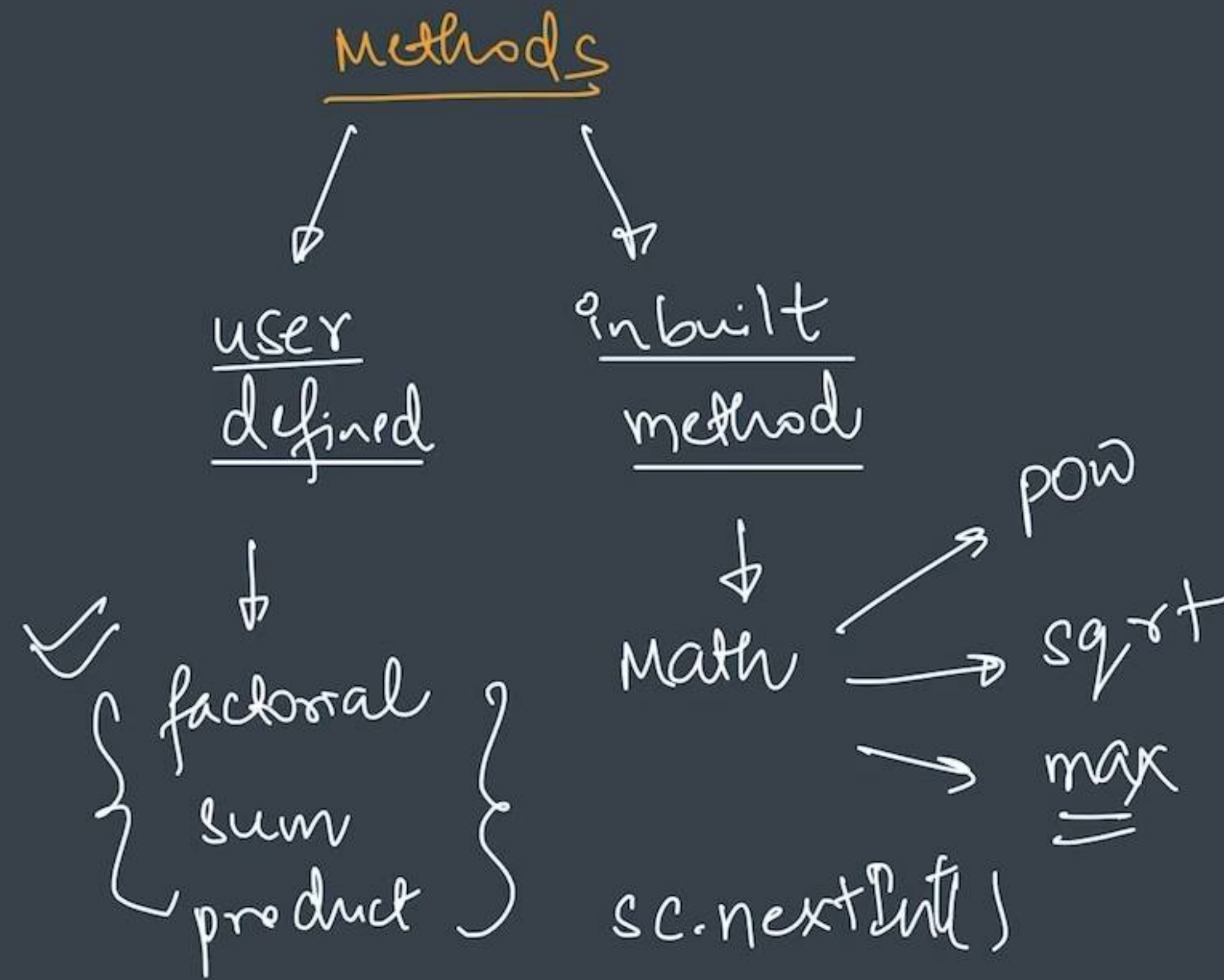
TERMINAL

```
shradhakhapra@Shradhas-Air Java Programming Language % java JavaBasics.java
```

10

```
shradhakhapra@Shradhas-Air Java Programming Language %
```





Function Overloading

1.2 x 5.6

double

Multiple functions with the same name but **different parameters**

Calculator

↓
multiply(int a, int b) 2 integer ✗ ✓
multiply 2 (float a, float b)
multiply 3 (double a, double b)

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multiply 4 (int a, int b,
int c)



double

Calculus

↓ multiply(int a, int b) ↘ X ✓

multiply ~~in~~ (float a, float b)

multiply ~~in~~ (double a, double b)

A circular video frame showing a woman with long dark hair speaking into a microphone. She is wearing a dark shirt. In the background, a lamp is visible on the left side. The video frame is set against a dark blue background.

Function Overloading

- type of param
- no. of params

1.2 x 5.6

double

Multiple functions with the same name but **different parameters**

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$\downarrow$
multiply (int a, int b,
int c)

Calculator

$\downarrow$
multiply (int a, int b) $\swarrow$ 2 integer $\times$ ✓
multiply (float a, float b) $\downarrow$
multiply (double a, double b) $\downarrow$



Function Overloading

using Parameters

f1: sum of 2 no.

```
int sum(int a, int b) {  
    return a + b;  
}
```

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f2: sum of 3 no.

```
int sum(int a, int b, int c) {  
    return a + b + c;  
}
```

int sum(int a, int b)

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float sum(int a, int b)



```
47 public static int sum(int a, int b) {  
48     return a + b;  
49 }  
50  
51 //func to calc sum of 3 nums  
52 public static int sum(int a, int b, int c) {  
53     return a + b + c;  
54 }
```

Run | Debug

```
55 public static void main(String args[]) {  
56     System.out.println(sum(3, 5));  
57     System.out.println(sum(5, 2, 1));  
58 }  
59 }
```

DEBUG CONSOLE

PROBLEMS

1

OUTPUT

TERMINAL

```
shradhakhapra@Shradhas-Air Java Programming Language % java JavaBasics.java
```

```
8
```

```
8
```

```
shradhakhapra@Shradhas-Air Java Programming Language %
```



Function Overloading

using Data Types

f1: add 2 int val

```
int sum(int a, int b) {  
    return a+b;  
}
```

f2: add 2 float val

```
float sum(float a, float b) {  
    return a+b;  
}
```



```
60
61 //func to cal float sum
62 public static float sum(float a, float b) {
63     return a+b;
64 }
```

Run | Debug

```
65 public static void main(String args[]) {
66     System.out.println(sum(3, 5));
67     System.out.println(sum(3.2f, 4.8f));
68 }
69 }
70
```

DEBUG CONSOLE

PROBLEMS

1

OUTPUT

TERMINAL

shradhakhapra@Shradhas-Air Java Programming Language % java JavaBasics.java

8

8.0

shradhakhapra@Shradhas-Air Java Programming Language %



Check if a number is Prime or not

① $n \rightarrow$



Check if a number is **Prime or not**

1) $n \rightarrow \text{prime} \rightarrow \underline{1, n}$

for (int i=2 to i=n-1)

↓
($n \% i == 0$) not prime

PRIME



```
66 public static boolean isPrime(int n) {
67     boolean isPrime = true;
68     for(int i=2; i<=n-1; i++) {
69         if(n % i == 0) { //completely dividing
70             isPrime = false;
71             break;
72         }
73     }
74
75     return isPrime;
76 }
```

Run | Debug

```
77 public static void main(String args[]) {
78     System.out.println(isPrime(12));
79 }
80 }
81
```

DEBUG CONSOLE

PROBLEMS

1

OUTPUT

TERMINAL

```
shradhakhapra@Shradhas-Air Java Programming Language % java JavaBasics.java
shradhakhapra@Shradhas-Air Java Programming Language % java JavaBasics.java
true
shradhakhapra@Shradhas-Air Java Programming Language % java JavaBasics.java
false
shradhakhapra@Shradhas-Air Java Programming Language %
```



```
65
66 public static boolean isPrime(int n) {
67     boolean isPrime = true;
68     for(int i=2; i<=n-1; i++) {
69         if(n % i == 0) { //completely dividing
70             isPrime = false;
71             return isPrime;
72         }
73     }
74
75     return isPrime;
76 }
```

Run | Debug

```
77 public static void main(String args[]) {
78     System.out.println(isPrime(12));
79 }
80 }
```

DEBUG CONSOLE

PROBLEMS

1

OUTPUT

TERMINAL

```
shradhakhapra@Shradhas-Air Java Programming Language % java JavaBasics.java
shradhakhapra@Shradhas-Air Java Programming Language % java JavaBasics.java
true
shradhakhapra@Shradhas-Air Java Programming Language % java JavaBasics.java
false
shradhakhapra@Shradhas-Air Java Programming Language %
```



```
65
66 public static boolean isPrime(int n) {
67     for(int i=2; i<=n-1; i++) {
68         if(n % i == 0) { //completely dividing
69             return false;
70         }
71     }
72
73     return true;
74 }
```

Run | Debug

```
75 public static void main(String args[]) {
76     System.out.println(isPrime(3));
77 }
78 }
79
```

DEBUG CONSOLE

PROBLEMS

1

OUTPUT

TERMINAL

```
shradhakhapra@Shradhas-Air Java Programming Language % java JavaBasics.java
false
shradhakhapra@Shradhas-Air Java Programming Language % java JavaBasics.java
true
shradhakhapra@Shradhas-Air Java Programming Language %
```



```

66 //only for n>=2
67 public static boolean isPrime(int n) {
68     //corner cases
69     //2
70     if(n == 2) {
71         return true;
72     }
73
74     for(int i=2; i<=n-1; i++) {
75         if(n % i == 0) { //completely dividing
76             return false;
77         }
78     }
79
80     return true;
81 }

```

Run | Debug

```

82 public static void main(String args[]) {

```

DEBUG CONSOLE PROBLEMS 1 OUTPUT TERMINAL

```

shradhakhapra@Shradhas-Air Java Programming Language % java JavaBasics.java
false
shradhakhapra@Shradhas-Air Java Programming Language % java JavaBasics.java
true
shradhakhapra@Shradhas-Air Java Programming Language % 

```



Check if a number is **Prime or not**

① $n \rightarrow \text{prime} \rightarrow \underline{1, n}$

for (int i=2 to i=n-1)

↓
 $(n \% i == 0)$ not prime

PRIME

optimized
 $6 \rightarrow 1 \times 6$



Check if a number is Prime or not

① $n \rightarrow \text{prime} \rightarrow \underline{1, n}$

for (int i=2 to i=n-1)

↓
($n \% i == 0$) not prime

PRIME

optimized

6 → 1x6
2x3
3x2
6x1

8 → 1x8
2x4
4x2
8x1

↓
for (int i=2 to $i \leq \sqrt{n}$)

$$n = \sqrt{n} \times \sqrt{n}$$

↓ ↑ ↑ ↓



```
76 //      return false;
77 //      }
78 //      }
79
80 //      return true;
81 // }
82
83 public static boolean isPrime(int n) {
84     if(n == 2) {
85         return true;
86     }
87     for(int i=2; i<=Math.sqrt(n); i++) {
88         if(n % i == 0) {
89             return false;
90         }
91     }
92 }
```

DEBUG CONSOLE PROBLEMS 1 OUTPUT TERMINAL

```
false
shradhakhapra@Shradhas-Air Java Programming Language % java JavaBasics.java
true
shradhakhapra@Shradhas-Air Java Programming Language % java JavaBasics.java
true
```



```
90         return false;
91     }
92 }
93
94     return true;
95 }
96
97
98 Run | Debug
99 public static void main(String args[]) {
100     System.out.println(isPrime(16));
101 }
102 }
```

DEBUG CONSOLE PROBLEMS 1 OUTPUT TERMINAL

```
shradhakhapra@Shradhas-Air Java Programming Language % java JavaBasics.java
true
shradhakhapra@Shradhas-Air Java Programming Language % java JavaBasics.java
false
shradhakhapra@Shradhas-Air Java Programming Language %
```



Print all Primes in a Range

$n = 10$

2 to 10

2, 3, 5, 7

```
ps void PrintInRange(int n) {  
    for (int i = 2 to n) {  
        isPrime(i) → True  
        → False  
    }  
}
```

isPrime(n)

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JavaBasics.java > JavaBasics > main(String[])

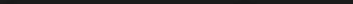
```

95     }
96
97     public static void primesInRange(int n) {
98         for(int i=2; i<=n; i++) {
99             if(isPrime(i)) { //true
100                 System.out.print(i+" ");
101             }
102         }

```

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DEBUG CONSOLE PROBLEMS 1 OUTPUT TERMINAL



```
shradhakhapra@Shradhas-Air Java Programming Language % java JavaBasics.java
2 3 5 7 11 13 17 19
shradhakhapra@Shradhas-Air Java Programming Language % java JavaBasics.java
2 3 5 7 11 13 17 19 23 29 31 37 41 43 47 53 59 61 67 71 73 79 83 89 97
shradhakhapra@Shradhas-Air Java Programming Language %
```



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Convert from Binary to Decimal $n = 101$

JAVA Programming Language



Convert from Binary to Decimal

$n = 101$

↓
Binary Number System

Bit Manipulation
↳ Bitwise Operators

0, 1



Convert from Binary to Decimal

n = 101

↓
Binary Number System

Math → 0 to 9

0, 1, 2, 3, 4, 5, 6, 7, 8, 9

↓
10 digits

Decimal Number System
↓
10



Convert from Binary to Decimal

n = 101

$$(0)_{10} = (0)_2$$

$$(1)_{10} = (01)_2$$

$$(2)_{10} = (10)_2$$

$$(3)_{10} = (11)_2$$

$$(4)_{10} = (100)_2$$

$$(5)_{10} = (101)_2$$

$$(6)_{10} = (110)_2$$

$$(7)_{10} = (111)_2$$

$$(8)_{10} = (1000)_2$$

⋮

⋮



Convert from Binary to Decimal

$n = 101$

↙ b_{1n}

$$\begin{array}{cccc} n = & 1 & 0 & 0 & 0 \\ & \downarrow & \downarrow & \downarrow & \downarrow \\ & 2^3 & 2^2 & 2^1 & 2^0 \end{array}$$

$$= 1(8) + 0(4) + 0(2) + 0(1)$$

$$= 8 + 0 + 0 + 0 = \underline{8}$$

$$(1000)_2 = (8)_{10}$$



Convert from Binary to Decimal

n = 101

↙ bin

- ① Paper & Pen
- ② logic
- ③ Code ✓

$$\begin{array}{cccc} n = & 1 & 0 & 0 & 0 \\ & \downarrow & \downarrow & \downarrow & \downarrow \\ & 2^3 & 2^2 & 2^1 & 2^0 \end{array}$$

$$= 1(8) + 0(4) + 0(2) + 0(1)$$

$$= 8 + 0 + 0 + 0 = \underline{8}$$

$$(1000)_2 = (8)_{10}$$

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Convert from Binary to Decimal

n = 101

n = 1 0 1 0 0 0 1 $\rightarrow$ 1 0 1 0 0 0 1

pow = 0

decimal = 0

$$\textcircled{1} \text{ dec} = \text{dec} + \text{LD} * 2^{\text{pow}} = 0 + 1(2^0)$$

LD = 1 pow++

$$\textcircled{2} \text{ dec} = \text{dec} + 1 * 2^1$$

$$= 1 + 1(2) = 3$$

- ① Paper & Pen
- ② logic
- ③ Code ✓



- ① Paper & Pen
 ② Logic
 ③ Code ✓

Convert from Binary to Decimal

n = 101

1 0 1
 ↓
 1 0 → 1

pow = ~~0~~ ~~1~~ 2
 dec = ~~0~~ 1
 LD = 1

$$\begin{aligned} \text{① } \text{dec} &= \text{dec} + [\text{LD} \times 2^{\text{pow}}] \\ &= 0 + [1 \times 2^0] = 0 + [1] = 1 \end{aligned}$$

$$\begin{aligned} \text{② } \text{dec} &= \text{dec} + [\text{LD} \times 2^{\text{pow}}] \\ &= 1 + [0 \times 2^1] = 1 + [0] = 1 \end{aligned}$$

$$\begin{aligned} \text{③ } \text{dec} &= \text{dec} + [\text{LD} \times 2^{\text{pow}}] = 1 + [1 \times 2^2] \\ &= 1 + 4 = 5 \end{aligned}$$



JavaBasics.java > JavaBasics > binToDec(int)

```
109     int pow = 0;
110     int decNum = 0;
111
112     while(binNum > 0) {
113         int lastDigit = binNum % 10;
114         decNum = decNum + (lastDigit * (int)Math.pow(2, pow));
115
116         pow++;
117         binNum = binNum/10;
118     }
119
120     System.out.println("decimal of " + myNum + " = " + decNum);
121 }
122
```

DEBUG CONSOLE

PROBLEMS

1

OUTPUT

TERMINAL

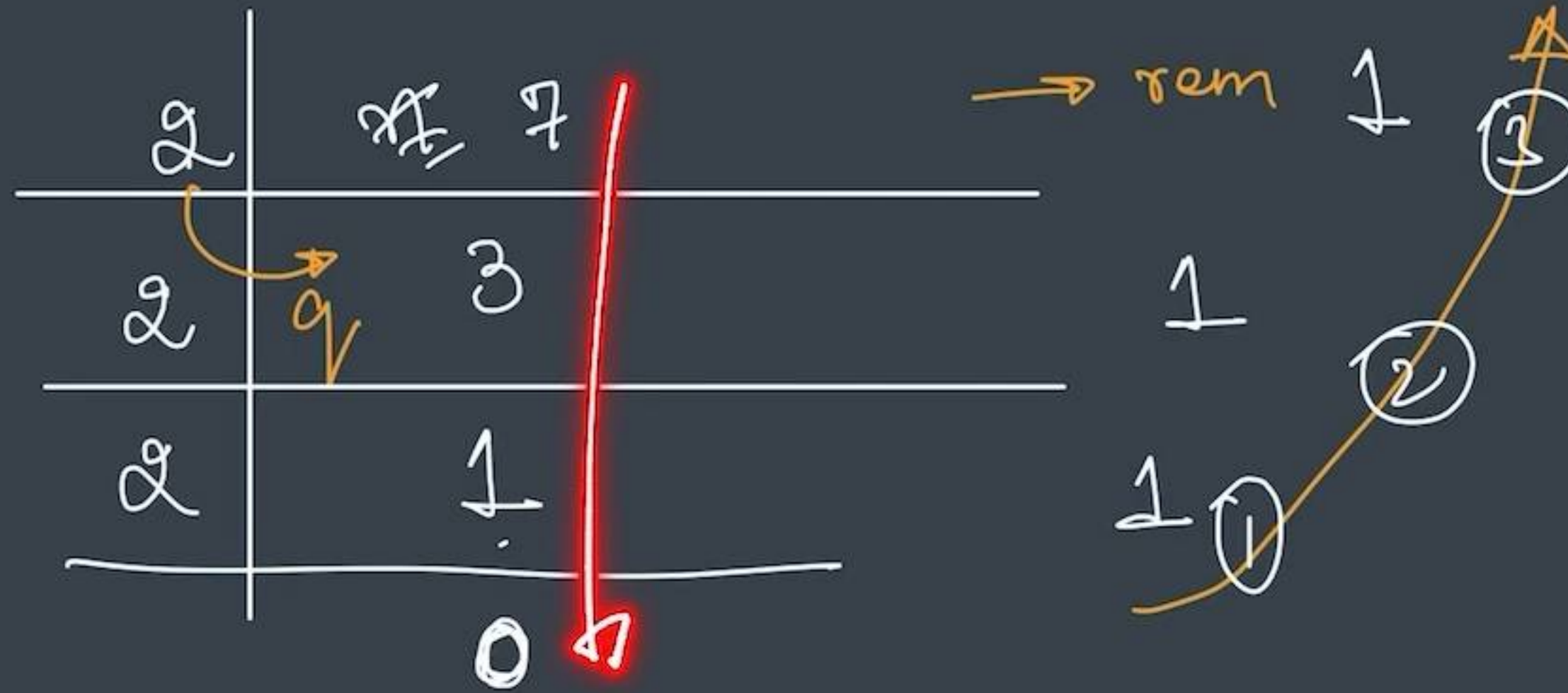
> java

```
shradhakhapra@Shradhas-Air Java Programming Language % java JavaBasics.java
```



Convert from **Decimal to Binary**

$n = 7$



$$7_{10} = \boxed{111}_2$$

Bin → Dec ✓

Dec → Bin ?

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1. P8P ✓

2. Logic

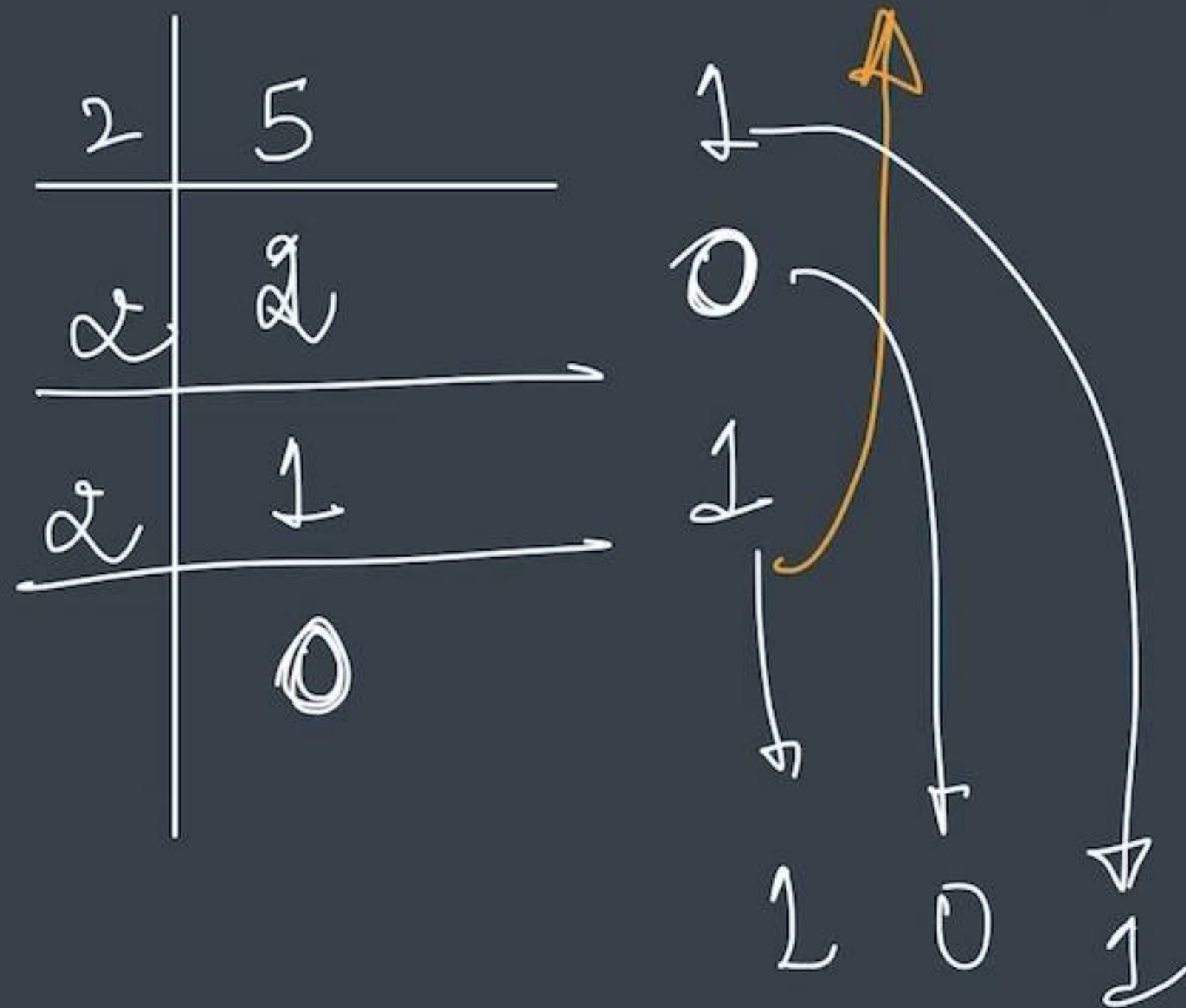
3. Code



Convert from Decimal to Binary

$n = 7$

$n = 5$



$$(101)_2 = (5)_{10}$$

Bin $\rightarrow$ Dec $\checkmark$

Dec $\rightarrow$ Bin ?

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1. P8P $\checkmark$

2. Logic

3. Code



Convert from Decimal to Binary n = 7

Bin $\rightarrow$ Dec $\checkmark$
Dec $\rightarrow$ Bin ? **APNA COLLEGE**

1. P8P $\checkmark$
2. Logic
3. Code

$n = 11$

2	11	1
2	5	1
2	2	0
2	1	1
	0	

0

$(1011)_2$

final no bin

while($n > 0$) {
 ① divide by 2
 $\rightarrow$ rem

$bin = bin + rem * 10^{pow}$
 $pow++$
}



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JavaBasics.java > JavaBasics > main(String[])

```
129         int rem = n % 2;
130         binNum = binNum + (rem * (int)Math.pow(10, pow));
131         pow++;
132         n = n/2;
133     }
134
135     System.out.println("binary form of " + myNum + " = " + binNum);
136 }
Run | Debug
137 public static void main(String args[]) {
138     decToBin(15);
139 }
140 }
141
```

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DEBUG CONSOLE

PROBLEMS

1

OUTPUT

TERMINAL

```
shradhakhapra@Shradhas-Air Java Programming Language % java JavaBasics.java
binary form of 7 = 111
shradhakhapra@Shradhas-Air Java Programming Language % java JavaBasics.java
binary form of 12 = 1100
shradhakhapra@Shradhas-Air Java Programming Language %
```



```
112     while(binNum > 0) {
113         int lastDigit = binNum % 10;
114         decNum = decNum + (lastDigit * (int)Math.pow(2, pow));
115
116         pow++;
117         binNum = binNum/10;
118     }
119
120     System.out.println("decimal of " + myNum + " = " + decNum);
121 }
122
123 public static void decToBin(int n) {
124     int myNum = n;
125     int pow = 0;
126     int binNum = 0;
127
128     while(n > 0) {
129         int rem = n % 2;
130         binNum = binNum + (rem * (int)Math.pow(10, pow));
131         pow++;
132         n = n/2;
133     }
134
135     System.out.println("binary form of " + myNum + " = " + binNum);
136 }
137
138 public static void main(String args[]) {
139     decToBin(15);
140 }
```

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Run | Debug



```
public static void binToDec(int binNum) {
    int myNum = binNum;
    int pow = 0;
    int decNum = 0;

    while(binNum > 0) {
        int lastDigit = binNum % 10;
        decNum = decNum + (lastDigit * (int)Math.pow(2, pow));

        pow++;
        binNum = binNum/10;
    }

    System.out.println("decimal of " + myNum + " = " + decNum);
}
```

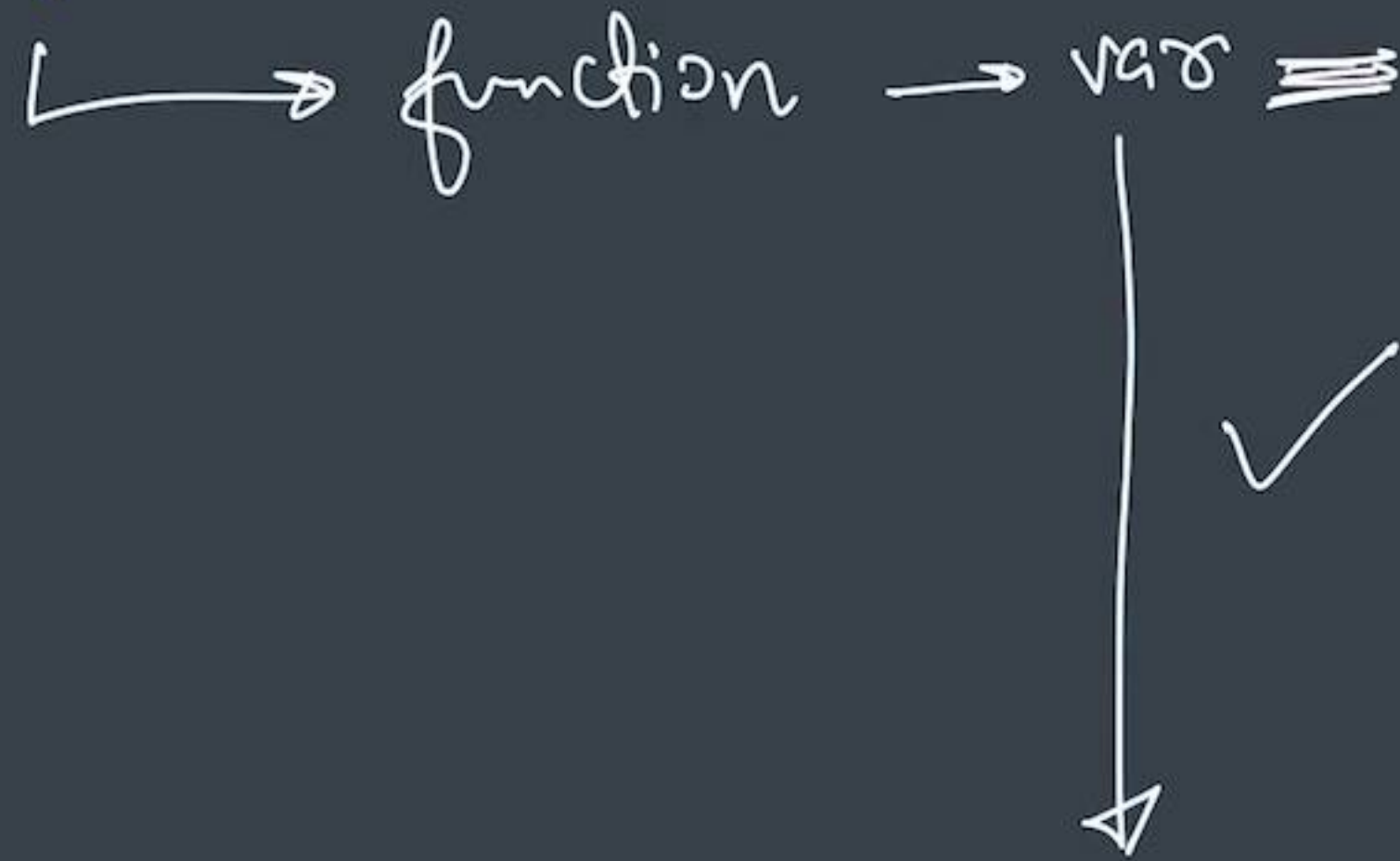
```
public static void decToBin(int n) {
    int myNum = n;
    int pow = 0;
    int binNum = 0;

    while(n > 0) {
        int rem = n % 2;
        binNum = binNum + (rem * (int)Math.pow(10, pow));
        pow++;
        n = n/2;
    }
}
```



Scope

Method Scope



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```
int sum(int a, int b) {  
    int s = a + b;  
}  
  
main() {  
    →  
}
```

Diagram illustrating method scope with handwritten annotations:

- An arrow points from the opening curly brace of the `sum` method to the variable `s`, indicating its scope is limited to that method. A checkmark is next to this arrow.
- Vertical lines separate the `sum` method from the `main` method.
- An arrow points from the opening curly brace of the `main` method to the right, indicating its scope.



JavaBasics.java > JavaBasics > main(String[])

```
129         int rem = n % 2;
130         binNum = binNum + (rem * (int)Math.pow(10, pow));
131         pow++;
132         n = n/2;
133     }
134
135     System.out.println("binary form of " + myNum + " = " + binNum);
136 }
137
138 public static void printS() {
139     int s = 45;
140 }
141 Run | Debug
142 public static void main(String args[]) {
143     System.out.println(s);
144 }
145 }
```



JavaBasics.java > JavaBasics > main(String[])

```
130      }
137
138      // public static void printS() {
139      //     int s = 45;
140      // }
141      Run | Debug
142      public static void main(String args[]) {
143      {
144      ..... int s = 45;
145      .....
146      }
147      System.out.println(s);
148      }
149  }
150
```



```
1360 // public static void printS() {
1361 //     int s = 45;
1362 // }
1363
1364 Run | Debug
1365
1366 public static void main(String args[]) {
1367     for(int i=1; i<=5; i++) {
1368
1369     }
1370     System.out.println(i);
1371 }
1372 }
1373
```

```
130
137
138 // public static void printS() {
139 //     int s = 45;
140 // }
141 Run | Debug
142 public static void main(String args[]) {
143     for(int i=1; i<=5; i++) {
144
145     }
146     System.out.println(i);
147 }
148
```



JavaBasics.java > JavaBasics > main(String[])

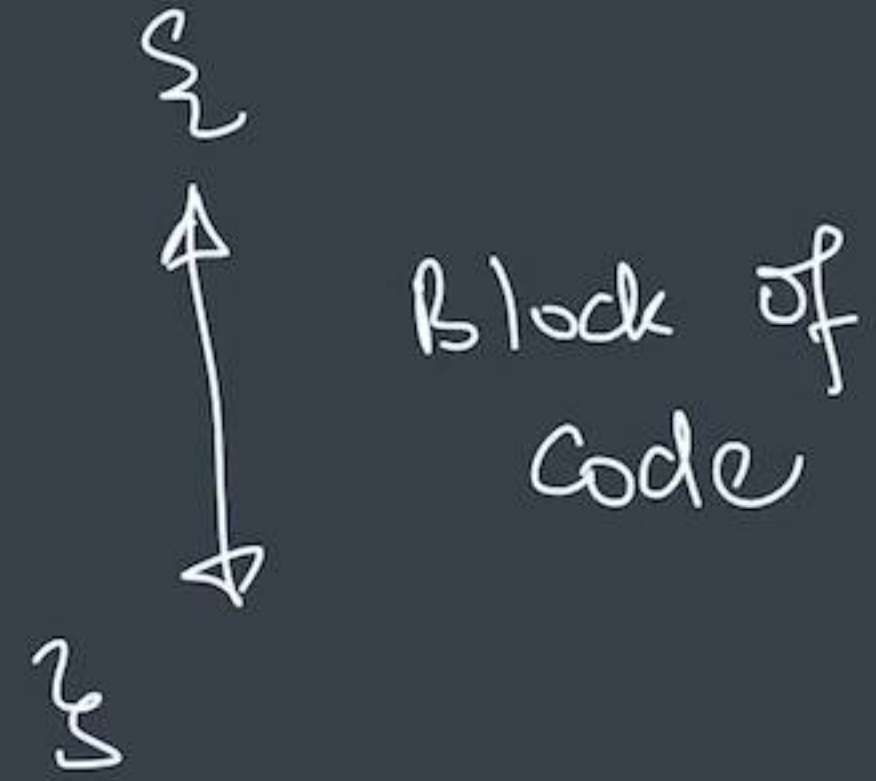
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```
130      }
137
138      // public static void printS() {
139      //     int s = 45;
140      // }
141      Run | Debug
142      public static void main(String args[]) {
143          for(int i=1, j=0, k=4; i<=5; i++) {
144              }
145          int i=4;
146          System.out.println(i); //4
147      }
148  }
149
```



Scope

Block Scope



class sc



Scope

Scope



Block of code

class scope ✓



access modifiers ✓



public ✓
private
protected }

OOP

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